

When the Margin Goes Negative: Sample Rejection as a Masked-Softmax Margin under Label Noise

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Abstract

Small-loss sample selection, the core heuristic of learning with noisy labels, stops training on examples the model currently fits poorly. We show that in the masked-softmax family of AS-Softmax this heuristic is a margin. A sample’s loss and gradient vanish exactly when its label gap $g = \max_{j \neq y} p_j - p_y$ satisfies $g \leq -\delta$, so a *negative* per-sample margin δ rejects the sample, and every selection rule is a choice of margin function. The view turns the rule’s suspicion statistic into the object of design and yields testable predictions. A batch-relative statistic is invariant to the batch’s overall fit, so the share it rejects cannot follow the noise rate. An absolute statistic (“another class beats the label”) rejects a closed-form band of gaps that does not depend on the noise rate, and leaves an escape region of confidently contradicted labels that no margin in the family can reject. We test these predictions in a pre-registered study: BERT-base on 20 Newsgroups, six synthetic noise conditions, five seeds, model selection on noisy held-out labels. The batch-relative margin ties the best robust loss at 40% symmetric noise but loses 3.8 points on clean data and 4.6 under pair-flip noise, as its noise-blindness predicts. The absolute margin memorizes the fewest corrupted labels in four of the five noisy conditions (8–10% under symmetric noise, against 13–29% for small-loss selection given the true noise rate) and is the best method under 40% pair-flip noise (+5.4 points over cross-entropy, +3.3 over oracle small-loss). It still trails GCE by 0.9–1.3 points on clean and symmetric noise, so it fails our pre-registered bar for a new method. Instrumented re-runs locate the costs. A rejected clean sample receives no gradient and stays locked out (23% of a noise-free training set under the batch-relative margin). AS-Softmax, the base loss, initially concentrates its gradient on mislabeled samples and memorizes them faster than cross-entropy. We report the result as a unifying view with measured consequences, not as a state-of-the-art method.

1 Introduction

Deep networks fit corrupted labels as readily as clean ones (Zhang et al., 2017), but they fit the clean ones first (Arpit et al., 2017; Liu et al., 2020). Two families of methods exploit this gap. *Robust losses* such as MAE (Ghosh et al., 2017), GCE (Zhang and Sabuncu, 2018) and SCE (Wang et al., 2019) bound the influence any single example can have on training. *Sample selection* methods such as MentorNet (Jiang et al., 2018), Co-teaching (Han et al., 2018) and DivideMix (Li et al., 2020) drop the examples the model currently fits worst, the *small-loss* trick. Selection is usually a procedure bolted onto the loss: rank the batch by loss, keep a fraction $1 - r$, and set r from an estimate of the noise rate, often its true value.

This paper starts from an observation about a different line of work. AS-Softmax (Lv et al., 2025) trains with cross-entropy over a *masked* softmax: a competing class j is dropped from the normalizer once the label beats it by a margin δ in probability, $p_y - p_j \geq \delta$. When every competitor is dropped, the sample’s loss and gradient are exactly zero. AS-Softmax uses $\delta > 0$, so this happens

only for samples the model already fits. Nothing in the mechanism requires $\delta > 0$, however. With $\delta < 0$ the same zero-loss region extends to samples whose label is *losing*, by up to $|\delta|$. A negative margin is therefore sample rejection (Proposition 1), and any selection rule, small-loss included, can be written as a per-sample margin function (Corollary 1).

The equivalence is simple, and its value lies in what it makes measurable. Once selection is a margin $\delta_i = \delta_b - \beta(2s_i - 1)$ driven by a per-sample suspicion statistic s_i , the statistic is the design choice and the margin’s geometry predicts behavior before any training:

- A **batch-relative** statistic, such as a z-score of p_y within the batch, is invariant to the batch’s overall level of fit (Proposition 3). The fraction it flags cannot follow the noise rate. The method then rejects about as much on clean data as on noisy data, and the strength β acts as a hidden, fixed rejection rate.
- An **absolute** statistic, $s_i = \sigma(g_i/\tau)$ with the label gap $g = \max_{j \neq y} p_j - p_y$, rejects a closed-form interval of gaps $[g_{lo}, g_{hi}]$ that depends only on (β, τ, δ_b) (Corollary 2). Nothing in it estimates or presumes the noise rate.
- Because the margin cannot fall below $\delta_b - \beta$, **no** statistic can reject a sample whose label is contradicted by more than $\beta - \delta_b$ (Proposition 2). Such samples are trained on against their leading class at full strength, and the same cap explains why memorization responds to β as a threshold rather than smoothly.

We test these predictions in a study whose hypotheses, pass criteria and decision rules were written down before each phase was run (Section 4). The study fine-tunes BERT-base (Devlin et al., 2019) on 20 Newsgroups (Lang, 1995). It covers symmetric noise at 0–60% and pair-flip noise at 20 and 40%, uses five seeds, and selects the model on held-out *noisy* labels, since a clean validation set is exactly what a practitioner facing label noise lacks. The predictions hold. The batch-relative margin has the highest mean accuracy of fourteen methods at the one condition its fixed strength happens to suit, 40% symmetric noise (a statistical tie with GCE). It costs 3.8 points on clean data and fails under pair-flip noise. The absolute margin removes most of the clean-data cost, memorizes the fewest corrupted labels in four of five noisy conditions and is the most accurate method under 40% pair-flip noise. It does *not* match GCE on clean or symmetric noise, and by our own pre-registered criterion it is not a new state-of-the-art method.

Contributions.

1. An exact equivalence between per-sample margins in masked softmax and sample rejection. Small-loss selection, disagreement filtering and AS-Softmax’s own sparsity then become points in one family of margin functions (Section 3).
2. Three consequences of the margin geometry: the noise-blindness of batch-relative statistics, the rate-free rejection band of the absolute statistic, and an escape region above $\beta - \delta_b$. Each is verified against the implementation in unit tests.
3. A pre-registered evaluation with five seeds, a realistic model-selection protocol and a tuning grid for the baselines. It reports every pre-registered comparison, including the ones the proposed rule loses (Section 5). Per-sample gradient instrumentation then locates each effect: rejection rates, the clean-data lockout and the escape region (Section 5.4).
4. Two negative findings: AS-Softmax memorizes corrupted labels *faster* than cross-entropy, and estimating the noise rate with a loss mixture, then rejecting, does worse here than rejecting disagreement directly.

2 Background and related work

Masked softmax. Let $o \in \mathbb{R}^C$ be the logits for an input with training label y (possibly corrupted) and $p = \text{softmax}(o)$. AS-Softmax (Lv et al., 2025) keeps the label and every competitor the label does not yet beat by a margin δ ,

$$K_\delta = \{y\} \cup \{j \neq y : \text{sg}(p_y - p_j) < \delta\}, \quad \ell_\delta(o, y) = -o_y + \log \sum_{j \in K_\delta} e^{o_j}, \quad (1)$$

where sg is stop-gradient: the mask is recomputed every step but no gradient flows through it. With $\delta = 1$ the mask keeps every class with $p_j > p_y - 1$, which is all of them unless $p_y = 1$ exactly, and ℓ_1 is cross-entropy. AS-Softmax uses a small positive constant ($\delta = 0.15$ here, reached after a 10% warm-up). It is one of many margin losses (Liu et al., 2016; Wang et al., 2018; Cao et al., 2019). It differs from the angular ones by working in probability space and by *removing* classes instead of shifting logits, which is what produces the exact zeros used below.

Sample selection. The small-loss trick trains on the fraction of each mini-batch with the lowest loss. The rejected fraction follows a schedule that ramps to a forget rate r , usually set to the true or estimated noise rate (Jiang et al., 2018; Han et al., 2018; Yu et al., 2019; Wei et al., 2020). DivideMix (Li et al., 2020), following Arazo et al. (2019), replaces the fixed rate with a two-component mixture fitted to per-sample losses, and Confident Learning (Northcutt et al., 2021) estimates per-class noise rates before pruning. A second line filters on *disagreement* between the prediction and the label: INCV (Chen et al., 2019) and SELF (Nguyen et al., 2020) keep samples whose (cross-validated or ensembled) prediction matches the label, SELFIE (Song et al., 2019) refurbishes labels with consistent predictions, and AUM (Pleiss et al., 2020) ranks samples by the margin between the assigned logit and the largest other one, averaged over training. Our absolute statistic is the probability-space, single-step version of this disagreement signal, applied inside the loss rather than as a separate filtering stage. Loss truncation (Kang and Hashimoto, 2020) applies the small-loss idea to text generation.

Selection as an implicit loss. Viewing selection as a loss is not new. Self-paced learning (Kumar et al., 2010) selects samples whose loss is below a threshold, and Meng et al. (2017) show its alternating optimization minimizes a concave, robust surrogate of the original loss. Trimmed loss minimization (Shen and Sanghavi, 2019) and the curriculum loss (Lyu and Tsang, 2020) make the same link for rank-based selection. These views act on the scalar *loss value*. Ours acts inside the softmax on the *label gap*. The selection threshold then has a direct geometric meaning (how far another class leads), and the AS-Softmax sparsity, small-loss rejection and disagreement filtering become one family whose members differ only in the statistic that sets δ_i .

Robust losses and label noise in text. Robust losses bound or reshape the gradient of poorly fitted samples instead of zeroing it: MAE (Ghosh et al., 2017), GCE (Zhang and Sabuncu, 2018), SCE (Wang et al., 2019), normalized losses (Ma et al., 2020). Others correct the targets (Reed et al., 2015; Patrini et al., 2017) or regularize toward early predictions (Liu et al., 2020). Surveys cover the field (Song et al., 2023), and CIFAR-N provides real human noise for images (Wei et al., 2022). For text classification, Jindal et al. (2019) add a noise-modelling layer, and Zhu et al. (2022) find pre-trained BERT fairly robust to injected noise when training is stopped early. Our protocol makes early stopping itself noisy by choosing the epoch on noisy held-out labels.

3 Rejection as a negative margin

Throughout, $g = \max_{j \neq y} p_j - p_y \in (-1, 1)$ is the *label gap*: $g < 0$ when the training label is the model’s top class, and $g > 0$ when some other class beats it. All statistics below are computed under stop-gradient, as the mask is. Proofs are in Section A. Every closed-form number quoted in this section is checked against the code in `tests/test_mechanism.py`.

3.1 The zero-loss region

Proposition 1 (Zero loss $\Leftrightarrow g \leq -\delta$). *For any $\delta \in [-1, 1]$, the masked loss (1) satisfies $\ell_\delta(o, y) = 0 \iff \nabla_o \ell_\delta(o, y) = 0 \iff g \leq -\delta$. If $g > -\delta$, the leading competitor $j^* = \arg \max_{j \neq y} p_j$ is in K_δ and $\ell_\delta > 0$.*

For $\delta > 0$ this is AS-Softmax’s sparsity: a sample stops contributing once its label wins by at least δ (*fitted*). For $\delta < 0$ the zero-loss region extends past $g = 0$: a sample whose label is beaten by up to $|\delta|$ also contributes nothing (*rejected*). Nothing in the loss distinguishes the two cases. The same exact zero that makes AS-Softmax sparse is the zero of a discarded sample.

Corollary 1 (Selection rules are margins). *Any per-sample keep/reject decision $a_i \in \{0, 1\}$ is realized by $\delta_i = 1$ if $a_i = 1$ and $\delta_i = -1$ otherwise: kept samples train with cross-entropy and rejected ones contribute nothing. In particular, small-loss selection with forget rate r is the binary margin $\delta_i = 1 - 2\mathbb{I}[\text{sample } i \text{ is among the } \lceil rB \rceil \text{ highest-loss samples of its batch}]$, up to the normalization of the batch mean.*

3.2 A family of suspicion margins

Small-loss uses two margin values. A smooth family interpolates between AS-Softmax and rejection.

Definition 1 (Suspicion margin). Given a per-sample suspicion statistic $s_i \in (0, 1)$ ($s_i > 1/2$: suspicious), a base margin δ_b , a ceiling $\delta_c \geq \delta_b$ and a strength $\beta(t) \geq 0$,

$$\delta_i = \text{clip}(\delta_b - \beta(t)(2s_i - 1), -1, \delta_c).$$

We use $\delta_b = \delta_c = 0.15$, the AS-Softmax value. The ceiling makes the family one-sided: a confident sample never trains under a tighter margin than AS-Softmax would give it. $\beta(t)$ ramps linearly from 0 to β over the first 30% of training and then holds, the shape of Co-teaching’s forget-rate schedule. $\beta = 0$ is AS-Softmax. We study three statistics:

- **batch-relative** (“M4”): $s_i = \sigma(z_i/T)$ with $z_i = (\bar{p}_y - p_{y,i})/\text{sd}_B(p_y)$, the z-score of the label probability within the mini-batch;
- **absolute gap**: $s_i = \sigma(g_i/\tau)$, suspicious iff another class currently beats the label;
- **loss mixture**: the posterior of the high-loss component of a two-component Gaussian mixture fitted to per-sample running log-losses, as in DivideMix (Section B).

Proposition 2 (Rejection condition and the escape region). *Under Definition 1, sample i has zero loss iff*

$$g_i \leq -\delta_c \quad (\text{fitted}) \quad \text{or} \quad \beta(2s_i - 1) \geq \delta_b + g_i \quad (\text{rejected}).$$

Consequently, a sample with $g_i \geq \beta - \delta_b$ is never rejected, whatever its statistic, and trains against at least its leading competitor j^ .*

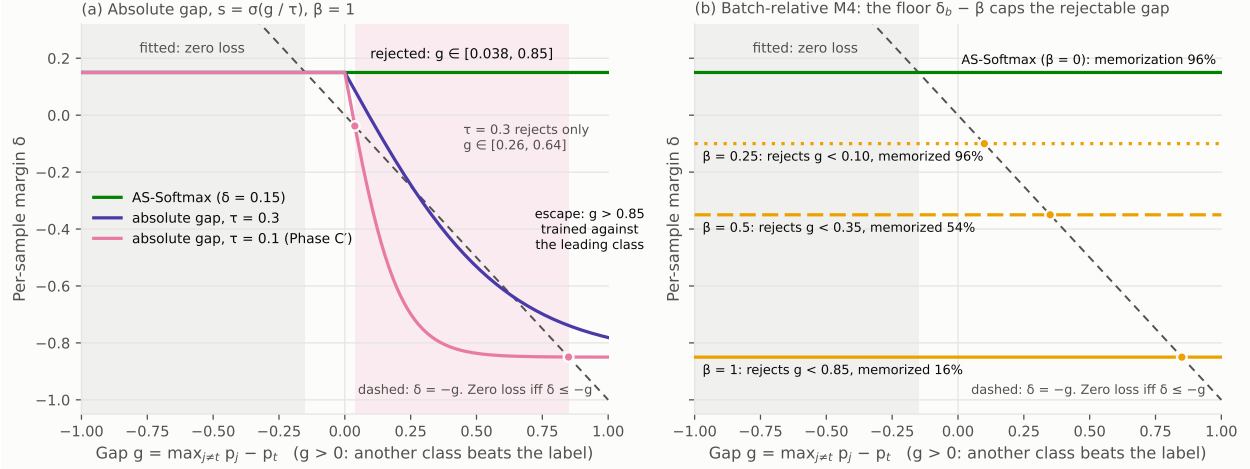


Figure 1: Margin geometry. A sample has zero loss iff its margin lies on or below the dashed line $\delta = -g$ (Proposition 1). (a) The absolute-gap margin rejects a fixed band of gaps, $[0.038, 0.85]$ at $\tau = 0.1$ and only $[0.26, 0.64]$ at $\tau = 0.3$, independent of the noise rate (Corollary 2). Above g_{hi} the sample escapes rejection. (b) The batch-relative margin can reach no lower than $\delta_b - \beta$, so it can only reject gaps below $\beta - \delta_b$ (Proposition 2). The memorization figures are from Table 1 (40% symmetric noise, last epoch).

The cap $\beta - \delta_b$ explains why memorization responds to β as a threshold (Figure 1b). With $\delta_b = 0.15$, strengths $\beta = 0.25, 0.5, 1$ can reject only samples whose label trails by less than 0.10, 0.35, 0.85. Once early learning has made the model confident in the true class of a mislabeled sample, its gap is large. A small β then rejects almost none of the noise. At 40% symmetric noise the three strengths end training with 96%, 54% and 16% of corrupted labels memorized; AS-Softmax ($\beta = 0$) reaches 96% (Table 1). The escape region also has a cost of its own. A confidently contradicted label, the clearest case of label noise, then receives a near-maximal per-sample gradient toward the wrong label. Under the absolute statistic below it is binary cross-entropy against j^* alone, with ℓ_1 -norm $2(1 - p_y / (p_y + p_{j^*})) \approx 2$.

3.3 Batch-relative statistics are blind to the noise rate

Proposition 3 (Affine invariance). *Let $s_i = h(z_i)$ for an increasing function h of the batch z -score z_i of $p_{y,i}$. Replacing every $p_{y,i}$ in the batch by a $p_{y,i} + b$ with $a > 0$ leaves every s_i unchanged. The fraction of the batch with $s_i \geq h(k)$, $k > 0$, is at most $1/(1 + k^2)$, and the fraction with $z_i > 0$ is the fraction below the batch mean, whatever the share of wrong labels in the batch.*

A batch with 40% wrong labels and a clean batch at an earlier stage of training can have the same standardized distribution of p_y , and then receive the same suspicion scores. The statistic measures *relative* fit only. It flags roughly the lower half of every batch, and how much of that half is rejected is set by β through Proposition 2, not by the data. The method therefore carries a fixed rejection rate that nobody chose explicitly. It suits one noise level and over- or under-rejects everywhere else. On clean data it discards correctly labeled samples that are merely below average. A rejected sample then receives no gradient and stays rejected, a lockout we measure in Section 5.4. Small-loss selection has the same blindness by construction. There the rate is an explicit input r , typically set to the true noise rate, which is information the method should not need.

3.4 An absolute statistic gives a rate-free rejection band

Corollary 2 (Rejection band). *With $s_i = \sigma(g_i/\tau)$ and full strength β , the rejected gaps are*

$$R = \{ g > 0 : \beta \tanh(g/2\tau) \geq \delta_b + g \},$$

an interval $[g_{lo}, g_{hi}]$ (possibly empty) that depends on (β, τ, δ_b) alone. For $\beta = 1$, $\delta_b = 0.15$: $R = [0.038, 0.850]$ at $\tau = 0.1$ and $R = [0.265, 0.635]$ at $\tau = 0.3$.

The band involves neither the batch, nor the noise rate, nor an estimate of either. What the rule rejects follows the data: on clean data, once the model fits, few samples have $g > 0$ and few are rejected; under heavy noise, many are. The rule is rate-free, not parameter-free. Its three constants fix the band in advance, and Figure 1a shows that the scale matters: at $\tau = 0.3$ the band shrinks to $[0.26, 0.64]$, the most contradicted labels escape, and memorization stays at the level of plain AS-Softmax (96%, Table 1). The absolute rule trains on two regions: near-ties, $g \in (-\delta_c, g_{lo})$, and the escape region $g > g_{hi}$.

3.5 AS-Softmax and memorization

Observation 1. Cross-entropy gives every sample a logit gradient of ℓ_1 -norm $2(1 - p_y) > 0$. AS-Softmax gives exactly zero to every fitted sample, $g \leq -\delta_b$. As clean samples become fitted, a larger share of each update comes from the samples that are not, and once early learning is over most of those carry wrong labels. Adam’s per-coordinate normalization (Kingma and Ba, 2015; Loshchilov and Hutter, 2019) makes the step size largely insensitive to the overall gradient scale. We therefore expect a larger gradient share on mislabeled samples, and faster memorization, while they are still unfitted. Once they are memorized, AS-Softmax counts them as fitted too and the effect reverses, with both losses eventually saturating.

The observation is a prediction, not a theorem. Section 5.4 measures both the memorization curves and the share of gradient carried by mislabeled samples.

4 Experimental protocol

Pre-registration. The study ran in phases. Before each phase we wrote down its hypotheses, numeric pass criteria and the decision each outcome would trigger, in a dated plan kept in the repository history (Section C). Phase A replicated an earlier two-seed result on five seeds. Phase B compared against GCE, SCE and small-loss selection; it failed as specified. Phase B’ removed two confounds named before it ran, and Phase C added noise types, noise rates, the strength sweep and the realistic selection protocol. Phase C’ tested the two self-calibrating statistics. Its rule read: if neither passes at its home condition, 40% symmetric noise, write the result up as a reframing and do not proceed to the planned real-noise benchmark. That is what happened, and this paper is the write-up. No method or hyperparameter was changed after its test results were seen. One design change was made before launch on a training-set diagnostic that involved no test accuracy (Section B).

Data and noise. 20 Newsgroups (Lang, 1995) with headers, footers and quotes removed, and empty documents dropped: 11,014 training and 7,317 test documents in 20 classes. We hold out 10% of the training set (1,101 documents) *with its noisy labels* as the model-selection set and train on the remaining 9,913. Test labels are never corrupted. *Symmetric* noise at rate η replaces each training label, independently with probability η , by a class drawn uniformly from the 19 others.

Pair-flip noise (Han et al., 2018) replaces it by $(y + 1) \bmod 20$, a systematic confusion the model can learn as a pattern. Conditions: symmetric $\eta \in \{0, 0.2, 0.4, 0.6\}$ and pair-flip $\eta \in \{0.2, 0.4\}$. The corruption is drawn once per condition and shared by every method and seed.

Model and training. `bert-base-uncased` (Devlin et al., 2019) with mean pooling and a linear head, inputs truncated to 128 tokens, dropout 0.1. AdamW (Loshchilov and Hutter, 2019) at a constant learning rate of 2×10^{-5} , weight decay 0.01, batch size 16, gradient clipping at 1.0, 8 epochs (4,960 steps), full precision. Seeds 42–46 vary the head’s initialization, dropout and batch order.

Model selection and metrics. Each run reports three test accuracies: **sel**, at the epoch with the highest accuracy on the noisy held-out labels (earliest on ties), our headline; **final**, at the last epoch, which is what training without any validation set yields; and **best**, at the epoch with the highest *clean test* accuracy, an oracle upper bound reported only for completeness. **Memorization** is measured on a fixed probe of 1,200 training examples, shared by all methods. Of the probe’s corrupted examples (477 at 40% symmetric noise), it is the fraction the model predicts as its corrupted label at the last epoch. For the mechanism analysis (Section 5.4) the probe also records, every epoch, each sample’s gap g and the ℓ_1 -norm of its logit gradient under the method’s own loss. That costs one autograd call on detached logits, with no backward pass through the model and no random draws, so training is unchanged.

Methods.

- **CE**: cross-entropy.
- **GCE** (Zhang and Sabuncu, 2018): $q = 0.7$; $q = 0.4$ in the tuning grid.
- **SCE** (Wang et al., 2019): $(\alpha, \beta) = (0.1, 1)$, $A = -4$; $(1, 1)$ in the grid.
- **Small-loss**: single-network small-loss selection, i.e. Co-teaching’s selection rule without the second network (Han et al., 2018). The forget rate ramps linearly to r over the first 30% of training, with $r = \eta$ (the *true* noise rate, an oracle) and $r = \eta/2$ in the grid.
- **AS-Softmax** (Lv et al., 2025): $\delta = 0.15$ with a 10% linear warm-up.
- **M4 (batch-relative)**: Definition 1 with the batch z-score, $T = 1$, $\beta \in \{0.25, 0.5, 1\}$ ($\beta = 1$ unless stated).
- **M4 absolute gap**: $\tau = 0.1$, $\beta = 1$; $\tau = 0.3$ as the one pre-registered scale alternative.
- **M4 loss mixture**: Section B.

Every suspicion margin uses $\delta_b = \delta_c = 0.15$, floor -1 , and the $0 \rightarrow 30\%$ ramp of β . The ramp matches small-loss’s schedule, a confound identified and removed in Phase B’. Baselines received the tuning grid above, evaluated with noisy-held-out selection at 40% symmetric noise. The grid changed GCE and SCE by less than 0.15 points. For small-loss, $r = \eta/2$ scored higher than the oracle rate (66.75 vs. 65.61, $t = 1.2$, Table 1). The grid ran only at 40% symmetric noise, so the multi-condition comparisons use the default settings listed first.

Statistics. We report mean $\pm$ standard deviation over the five seeds, and paired differences between methods over the same seeds with the paired t statistic (4 degrees of freedom; $|t| > 2.78$ corresponds to $p < 0.05$, two-sided). With five seeds we call a difference *established* only past that threshold. A run takes about 11 minutes on one RTX 3080 Ti; the study used about 280.

5 Results

5.1 The home condition: 40% symmetric noise

Table 1 gives the condition the batch-relative margin was designed around. Four observations stand out.

Model selection on noisy labels compresses everything. Under sel, twelve of the fourteen methods fall within 0.8 points of CE, and none is established as better than CE. Early stopping on noisy held-out labels already captures most of what robust training buys at this noise rate. The differences reappear without selection: at the last epoch CE has dropped to 51.5%, and the two rejection margins hold 66.7% (batch-relative) and 65.9% (absolute), above GCE (65.6) and small-loss (64.4).

The absolute margin memorizes least. It ends training having fitted 9.5% of the corrupted probe labels. That is the lowest of all fourteen methods, against 24–26% for GCE and oracle small-loss and 96% for CE. It does so without being told the noise rate.

The strength sweep is a threshold, as Proposition 2 predicts. Memorization goes 96 / 96 / 54 / 16% for $\beta = 0/0.25/0.5/1$. The cap $\beta - \delta_b$ explains the shape. At $\beta = 0.25$ only gaps below 0.10 are rejectable, while the mislabeled samples’ gaps are large once early learning has happened. The same geometry predicts the failure of the absolute gap at $\tau = 0.3$. Its band $[0.26, 0.64]$ lets the most contradicted labels escape, and it memorizes like AS-Softmax (95.6%).

Estimating the rate is not enough. The loss-mixture statistic estimates the noise rate well (0.35 ± 0.01 against a true 0.40) but is the least accurate margin, 1.64 points below CE ($t = -3.0$). It also memorizes more than the absolute rule (23% vs. 9.5%). Knowing *how many* labels are wrong did not tell it *which*.

5.2 Across noise conditions

Table 2 and Figures 2 and 3 test the two statistics’ predictions away from the home condition.

The batch-relative margin is noise-blind, as Proposition 3 predicts. On clean data it loses 3.83 points to CE (5/5 seeds, $t = -5.2$), and 6.8 at the last epoch: it keeps discarding correctly labeled samples that are merely below their batch’s average. At 60% symmetric noise it under-rejects (memorization 33%, twice its 40% level) and trails GCE by 2.0 points ($t = -3.4$). Under 40% pair-flip noise it is the worst method, 4.6 points below CE ($t = -2.0$) and 6.7 below small-loss ($t = -3.6$). The strength that suits 40% symmetric noise is a hidden rejection rate that suits nothing else.

The absolute margin follows the data, but pays on easy regimes. Its clean-data cost falls from 3.83 to 1.51 points; that is still established ($t = -5.1$), so P1 fails. Under structured noise it is the best method: at 40% pair-flip it reaches 62.69, +5.38 over CE ($t = 3.4$, 5/5 seeds) and +3.27 over small-loss given the true rate ($t = 3.3$). It is +4.93 over GCE, though GCE’s variance across seeds is too large to establish that ($t = 1.7$). Its memorization there (26%) is half of small-loss’s and a third of GCE’s. On symmetric noise it trails GCE by 1.0 ($\eta = 0.2$, $t = -3.4$), 0.95 ($\eta = 0.4$, $t = -1.9$) and 1.3 points ($\eta = 0.6$, $t = -2.1$). The rule rejects every sample the model currently disagrees with. That includes hard but correctly labeled samples, and on easy regimes GCE’s soft down-weighting is the better trade.

Table 1: 40% symmetric noise, all fourteen methods (5 seeds, mean $\pm$ sd). **sel**: test accuracy at the epoch chosen on noisy held-out labels (headline). **final**: last epoch. **best**: oracle epoch chosen on the clean test set. **mem**: corrupted probe labels predicted as given at the last epoch. Δ CE: paired sel difference to cross-entropy, with t .

	method	sel	final	best (oracle)	mem	Δ CE sel (t)
baselines	CE	66.97 $\pm$ 0.48	51.51 $\pm$ 1.22	67.08 $\pm$ 0.50	96.1	—
	GCE, $q = 0.7$	67.34 $\pm$ 0.78	65.55 $\pm$ 0.86	67.80 $\pm$ 0.34	24.3	+0.36 (0.7)
	GCE, $q = 0.4$	67.20 $\pm$ 0.60	57.51 $\pm$ 1.41	67.62 $\pm$ 0.45	60.6	+0.23 (0.6)
	SCE, (0.1, 1)	66.64 $\pm$ 0.88	63.07 $\pm$ 0.75	67.69 $\pm$ 0.28	29.1	−0.33 (−0.7)
	SCE, (1, 1)	66.73 $\pm$ 0.69	54.62 $\pm$ 1.16	67.50 $\pm$ 0.56	75.6	−0.24 (−0.6)
	small-loss, $r = \eta$ (oracle)	65.61 $\pm$ 1.19	64.38 $\pm$ 0.94	67.10 $\pm$ 0.33	25.9	−1.37 (−2.3)
	small-loss, $r = \eta/2$	66.75 $\pm$ 1.13	58.70 $\pm$ 0.63	67.11 $\pm$ 0.57	58.2	−0.22 (−0.5)
batch-rel.	AS-Softmax ($\beta = 0$)	66.71 $\pm$ 0.91	53.42 $\pm$ 0.64	67.40 $\pm$ 0.47	96.4	−0.26 (−0.7)
	M4, $\beta = 0.25$	66.74 $\pm$ 0.99	53.04 $\pm$ 2.74	67.43 $\pm$ 0.54	96.0	−0.24 (−0.4)
	M4, $\beta = 0.5$	66.84 $\pm$ 0.63	58.49 $\pm$ 1.47	67.19 $\pm$ 0.54	53.8	−0.14 (−0.5)
	M4, $\beta = 1$	67.44 $\pm$ 0.55	66.72 $\pm$ 0.80	67.64 $\pm$ 0.28	16.1	+0.47 (1.4)
self-calib.	absolute gap, $\tau = 0.1$	66.39 $\pm$ 0.65	65.93 $\pm$ 1.01	67.01 $\pm$ 0.44	9.5	−0.59 (−1.8)
	absolute gap, $\tau = 0.3$	66.56 $\pm$ 1.15	52.58 $\pm$ 1.57	67.23 $\pm$ 0.49	95.6	−0.41 (−0.8)
	loss mixture	65.33 $\pm$ 0.90	64.55 $\pm$ 0.34	66.26 $\pm$ 0.53	23.1	−1.64 (−3.0)

Without a validation set, the absolute margin’s advantage grows with the noise. At the last epoch its difference to GCE goes from -2.1 (clean) and -1.3 (20%) to $+0.4$ (40%), $+5.1$ (60%, $t = 9.7$, 5/5 seeds) and $+7.1$ (pair 40%, $t = 3.1$). Under symmetric noise the absolute margin loses at most 0.9 points between the selected and the last epoch (GCE: up to 7.3), because it keeps refusing to fit what it disagrees with. Under pair-flip noise both decay more (absolute gap 6.3, GCE 8.4 points at 40%).

5.3 Pre-registered verdicts

Table 3 collects the pre-registered tests. The decisive one is P3. The bar was the best tuned baseline at the home condition minus half a point, $67.34 - 0.5 = 66.84$, and neither self-calibrating statistic reached it. By the rule we fixed in advance, the absolute-gap margin is therefore not presented as a new method, and the real-noise benchmark (CIFAR-10N/100N, Wei et al., 2022) was not run. P2 failed for a reason we only understood afterwards. Its measure, the fraction of masked non-target slots, is dominated by AS-Softmax’s masking of fitted samples (0.996 for every method in the family at every noise rate), so it cannot register sample-level adaptivity. The sample-level evidence is the flat 8–10% memorization across symmetric noise rates in Table 2, against small-loss’s 29% $\rightarrow$ 13%. We report P2 as failed, as worded.

5.4 Mechanism

AS-Softmax memorizes faster than cross-entropy. Observation 1 predicts that zeroing the gradient of fitted samples shifts each update toward the unfitted, mostly mislabeled ones. Figure 4 confirms the prediction. AS-Softmax’s memorization exceeds CE’s at epochs 3 through 7 on all five seeds: by 2.5 points at epoch 3 ($t = 3.5$), 7.2 at epoch 4 ($t = 5.5$), 9.2 at epoch 5 ($t = 13.0$), 6.5 at epoch 6 and 3.5 at epoch 7. Both saturate at 96% by epoch 8. The same holds in the earlier 4-epoch runs (39.4% vs. 34.7% at the last epoch). On its own, the base loss of the margin family is *less* noise-robust than cross-entropy. As a base for rejection it is nonetheless the better choice. In a

Table 2: All six conditions for the five main methods (5 seeds, mean $\pm$ sd). Top: test accuracy with noisy-held-out selection. Middle: last epoch. Bottom: memorization of corrupted labels at the last epoch (%). Small-loss equals CE at $\eta = 0$ ($r = 0$). Small-loss is given the true rate.

method	symmetric				pair-flip	
	0%	20%	40%	60%	20%	40%
<i>sel (noisy held-out selection)</i>						
CE	71.38 ± 0.61	68.34 ± 0.80	66.97 ± 0.48	63.65 ± 0.99	67.39 ± 1.48	57.31 ± 4.79
small-loss	71.38 ± 0.61	68.72 ± 0.83	65.61 ± 1.19	63.41 ± 1.24	67.62 ± 0.64	59.42 ± 2.35
GCE	71.15 ± 0.39	68.86 ± 0.27	67.34 ± 0.78	64.55 ± 0.51	68.16 ± 0.52	57.76 ± 6.15
M4 (batch-rel.)	67.55 ± 1.27	68.49 ± 0.51	67.44 ± 0.55	62.52 ± 1.37	66.93 ± 0.66	52.71 ± 3.65
absolute gap	69.88 ± 0.37	67.83 ± 0.66	66.39 ± 0.65	63.27 ± 1.03	68.04 ± 0.52	62.69 ± 1.55
<i>final (last epoch)</i>						
CE	71.25 ± 0.14	62.94 ± 0.68	51.51 ± 1.22	36.89 ± 1.57	61.96 ± 1.37	48.95 ± 1.39
small-loss	71.25 ± 0.14	67.39 ± 0.91	64.38 ± 0.94	61.99 ± 0.39	65.33 ± 0.87	53.63 ± 2.61
GCE	70.85 ± 0.52	68.91 ± 0.33	65.55 ± 0.86	57.24 ± 0.60	67.35 ± 0.78	49.32 ± 0.71
M4 (batch-rel.)	64.43 ± 2.25	68.51 ± 0.55	66.72 ± 0.80	54.92 ± 2.07	64.98 ± 0.70	47.19 ± 1.85
absolute gap	68.75 ± 0.89	67.60 ± 0.90	65.93 ± 1.01	62.38 ± 1.55	66.43 ± 1.18	56.44 ± 5.13
<i>memorization at the last epoch (%)</i>						
CE	—	93.5	96.1	97.4	96.6	97.1
small-loss	—	29.4	25.9	13.4	50.9	48.5
GCE	—	16.1	24.3	31.4	39.3	80.4
M4 (batch-rel.)	—	7.0	16.1	32.6	34.3	66.5
absolute gap	—	9.3	9.5	8.0	17.5	26.3

Phase B' control with identical rejection decisions, training the kept samples with CE instead of AS-Softmax cost 4.1 points (63.3 vs. 67.4, 4 epochs, Section C). We have not isolated why.

Where the gradient goes. We re-ran six methods at 40% symmetric noise, three without noise and five at 40% pair-flip, on three seeds, recording each probe sample's gap and logit gradient every epoch (Figure 5). The instrumentation reads the logits only, and the re-runs reproduce the Phase C accuracies exactly. Four findings stand out.

AS-Softmax's faster memorization follows its gradient share. At epochs 1–4 the share of AS-Softmax's gradient carried by mislabeled samples exceeds CE's by 8.5, 11.6, 13.7 and 11.8 points (3/3 seeds, $t \geq 11$). This is the period in which its memorization pulls ahead. Once the mislabeled samples are memorized, AS-Softmax counts them as fitted and stops pushing on them, and from epoch 6 its share falls below CE's (by 13 to 29 points). This is Observation 1 in its precise form.

The rejection margins reject the right samples, at different rates. From epoch 2 on, the absolute rule gives zero gradient to 89–93% of the mislabeled probe samples, the batch-relative rule to 78–85%, and small-loss given the true rate to 72–83%. Under 40% pair-flip noise the gaps widen: 73–81%, 32–36% and 48–64%. The batch-relative rule rejects only a third of the pair-flipped labels, which is the mechanism behind its pair-flip failure.

The clean-data cost is a lockout. Without any noise, the batch-relative rule rejects 14% of the (correctly labeled) training samples after one epoch and 23% at the end. Its training accuracy stays between 76 and 79% for all eight epochs, against 99.7% for CE. A rejected sample receives no gradient, so nothing moves it back into the trained region. The absolute rule locks out 13–15% and reaches 86% training accuracy, in line with its smaller clean-data cost (1.5 vs. 3.8 points).

The escape region is sparsely populated but expensive. Under the absolute rule, at most 6%

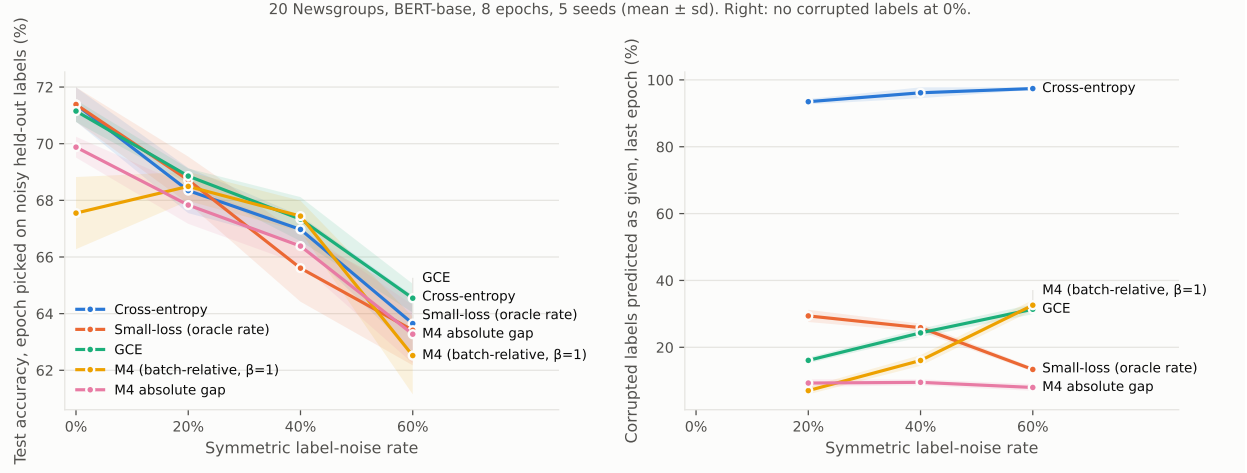


Figure 2: Symmetric noise. Left: test accuracy with noisy-held-out selection. Right: memorization of corrupted labels at the last epoch. Mean $\pm$ sd over 5 seeds. The batch-relative margin (yellow) is competitive only near 40%, the rate its fixed strength suits. The absolute margin (pink) keeps memorization at 8–10% at every rate.

of mislabeled probe samples ever reach $g \geq 0.8$ (4% under pair-flip noise), because models built on AS-Softmax stay unconfident. Under GCE and small-loss, 57–75% do. Yet nearly every other sample has zero gradient, so those few carry 10–35% (mean 20%) of the probe’s entire gradient at epochs 4–8, all of it pushing toward the wrong label; the variation across seeds is large. The untested remedy $\beta \geq 1 + \delta_b$ (Section 6) would remove exactly this gradient.

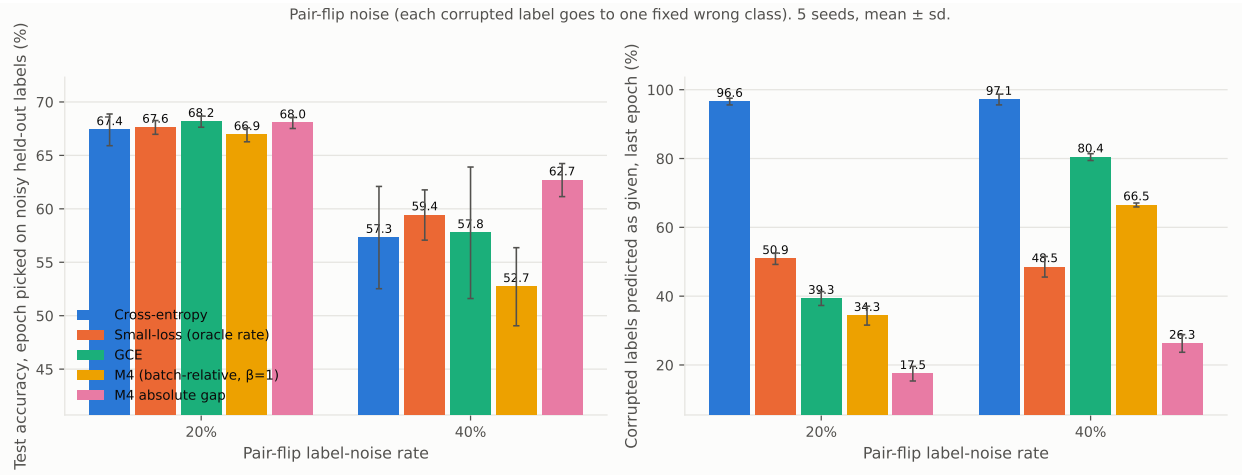


Figure 3: Pair-flip noise, where each corrupted label goes to one fixed wrong class. The absolute margin is the only method whose accuracy holds at 40% and the only one that keeps memorization well below half.

Table 3: Pre-registered criteria (fixed before the corresponding runs) and outcomes. Accuracies use sel unless noted. N1–N4 were defined for the batch-relative M4 (Phases A–C), P1–P4 for the self-calibrating statistics (Phase C'). The loss mixture was eliminated at P3 and, by the pre-registered rule, not run elsewhere.

criterion	outcome	verdict
N1 replicate (4 ep., final)	M4 – CE = +1.81, $t = 4.4$, 5/5 seeds; memorization 17% vs. 35%	pass
N2 $\geq$ small-loss – 0.5	4 ep. final: –0.39 after removing the timing confound (B'); 8 ep. sel: +1.84	pass
N3 $\geq$ max(GCE, SCE)	4 ep. final: –0.49 vs. GCE; 8 ep. sel: +0.11	fail at 4 ep., tie at 8
N4 memorization falls steadily in β	96 / 96 / 54 / 16%: monotone, threshold-shaped (Proposition 2)	partial
P1 clean cost $\geq$ CE – 0.5	absolute gap –1.51 ($t = -5.1$)	fail
P2 adaptivity	masked-slot ratio flat (0.996) at every rate; mixture estimate 0.35 at $\eta = 0.4$	fail as worded
P3 home $\geq$ best baseline – 0.5 = 66.84	absolute gap 66.39; mixture 65.33	fail (both)
P4 pair 40%: $\geq$ CE, within 2 of small-loss	absolute gap 62.69 vs. 57.31 / 59.42	pass

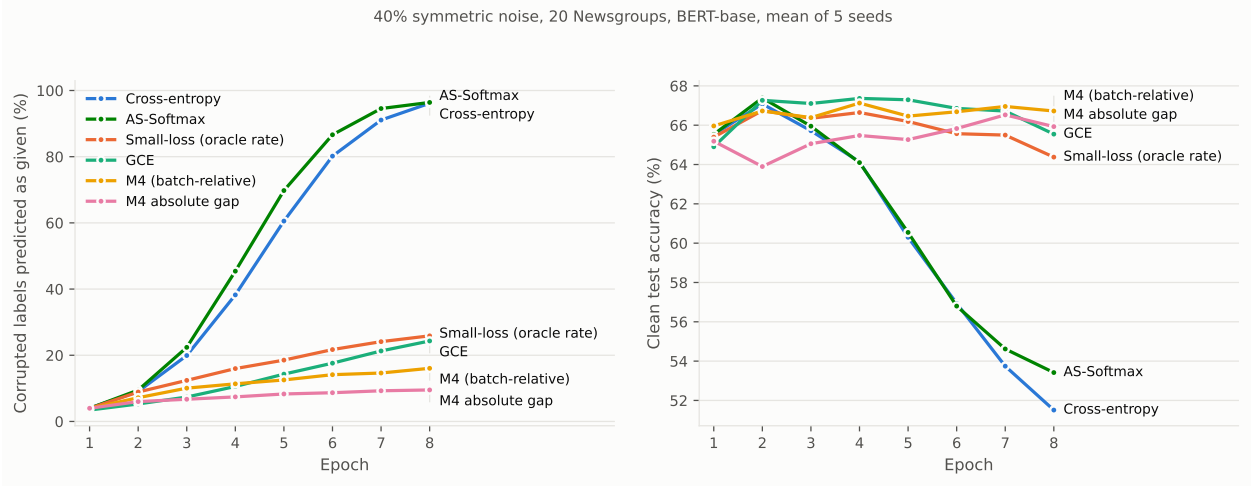


Figure 4: 40% symmetric noise, per epoch, mean of 5 seeds. Left: memorization of corrupted labels. AS-Softmax (green) memorizes faster than cross-entropy at epochs 3–7, on every seed. Right: clean test accuracy, the degradation that noisy-held-out selection has to catch.

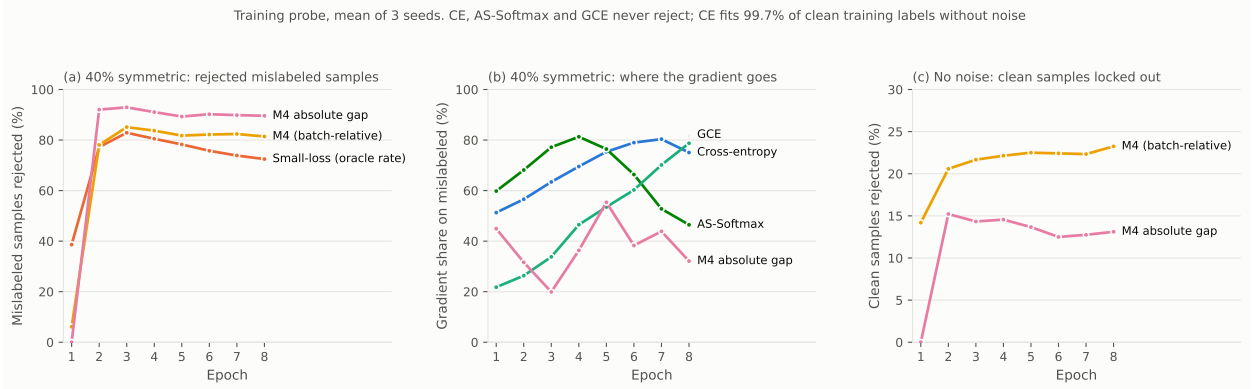


Figure 5: Mechanism, read every epoch on the training probe (mean of 3 seeds). (a) Fraction of mislabeled samples that are rejected: zero gradient although another class beats the label. At epoch 1 the absolute rule rejects nothing, because β has only ramped to 0.42 and its band (Corollary 2) is still empty. (b) Share of the probe’s gradient carried by mislabeled samples. AS-Softmax exceeds CE until its mislabeled samples are memorized and count as fitted. (c) With no noise at all, the rejection rules lock clean samples out of training.

6 Discussion

What the margin view buys. Writing rejection as a margin did not produce a method that beats GCE on symmetric noise. It did make three properties of selection rules visible before any training run, and each was then measured. First, *relative* statistics carry a hidden rate. A z-score, a rank or a fixed budget rejects the same share of a clean batch as of a corrupted one (Proposition 3). That is why the batch-relative margin wins exactly at one noise level and why small-loss needs the oracle rate. Second, an *absolute* statistic turns the rejection threshold into a constant of the loss, a band of label gaps (Corollary 2). The rejected share then follows the data, including structured noise, where rate-based selection has no reason to pick the right samples. Third, any margin family with a floor has an *escape region* (Proposition 2). It explains the threshold-shaped strength sweep and the failure of the wider-temperature variant. It also identifies a defect of our recipe: the most confidently contradicted labels receive a near-maximal gradient toward the wrong class. The measured footprint is small in population but not in gradient. At most 6% of mislabeled samples reach the region, yet they carry about a fifth of the gradient late in training (Section 5.4). The remedy is $\beta \geq 1 + \delta_b$, which makes the band reach $g = 1$; with $\beta = 1.15$, $\tau = 0.1$ it becomes $[0.032, 1.0)$. We did not run it, because the pre-registered protocol froze β before Phase C', and we leave it as a stated prediction.

When to reject and when to down-weight. The absolute rule discards every sample the model currently disagrees with. Under symmetric noise many of those are hard but correct, and GCE, which only down-weights them, is 0.9–1.3 points better. Under pair-flip noise, disagreement is the better signal. A fixed wrong class is learned as a pattern, so mislabeled samples do not stand out by loss rank (small-loss memorizes half of them). They appear to stand out by disagreement before the pattern is learned (the absolute rule memorizes a quarter). The same contrast appears without model selection. The heavier or more structured the noise, the larger the absolute rule's last-epoch lead over GCE: +5.1 at 60% symmetric and +7.1 at 40% pair-flip. Rejection also has a failure mode that down-weighting lacks. A rejected sample gets no gradient, so a wrongly rejected clean sample is locked out for good: 23% of a noise-free training set under the batch-relative rule, 13% under the absolute one. Readmitting rejected samples, for instance by periodically relaxing β , is the obvious remedy; we did not test it.

Limitations. One dataset, one model, one modality, and synthetic noise only. The real-noise benchmark was cut by our own pre-registered rule, so the claims above are about 20 Newsgroups with injected noise. Five seeds resolve differences of about one point at best: several comparisons we describe (+4.9 over GCE under pair-flip, -0.95 at 40% symmetric) are not established at $p < 0.05$, and the mechanism readings rest on three seeds and a 1,200-sample probe. Small-loss is the single-network rule, not Co-teaching's two networks, and DivideMix-style semi-supervised pipelines are out of scope, since they are training procedures rather than losses. The baselines' tuning grid ran only at 40% symmetric noise. Training was fixed at 8 epochs with a constant learning rate. The suspicion margins inherit AS-Softmax's base margin, which we showed memorizes faster than CE on its own.

7 Conclusion

In masked softmax, a sample's loss vanishes exactly when its label gap reaches the negative of its margin. A negative margin is therefore sample rejection, and the choice of selection rule becomes the choice of a margin function. The view makes falsifiable predictions, and they held in a pre-registered

study. Relative statistics are noise-blind. An absolute “another class beats the label” statistic rejects a fixed, rate-free band of gaps, memorizes the fewest corrupted labels and is the most robust rule under structured noise. A margin floor leaves an escape region that shapes how the rule responds to its strength, and rejection without readmission locks hard clean samples out of training. The absolute rule did not clear our bar against GCE on clean and symmetric noise, and we report it as a unifying account with measured consequences, not as a new state of the art.

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A Proofs

Proposition 1. Every $j \neq y$ is dropped iff $p_y - p_j \geq \delta$ for all $j \neq y$, i.e. iff $\min_{j \neq y} (p_y - p_j) = -g \geq \delta$. Then $K_\delta = \{y\}$ and $\ell_\delta = -o_y + \log e^{o_y} = 0$. Because the mask is under stop-gradient it is held fixed when differentiating, and the loss is identically zero as a function of o , so $\nabla_o \ell_\delta = 0$. Otherwise $K_\delta \supsetneq \{y\}$. Writing $\tilde{p}$ for the softmax restricted to K_δ , $\ell_\delta = -\log \tilde{p}_y > 0$ and $\partial \ell_\delta / \partial o_y = \tilde{p}_y - 1 \neq 0$. Finally, $p_y - p_{j^*} = -g < \delta$, so $j^* \in K_\delta$. $\square$

Corollary 1. $\delta = -1$ gives $-g \geq -1 = \delta$ for every sample, so by Proposition 1 the loss and gradient vanish. $\delta = 1$ keeps every j with $p_j > p_y - 1$, which is every class unless $p_y = 1$ exactly, so ℓ_1 is cross-entropy. The batch mean then averages over all B samples rather than the kept ones, a constant factor per batch. $\square$

Proposition 2. Write $u = \delta_b - \beta(2s - 1)$, so $\delta = \min(\max(u, -1), \delta_c)$. By Proposition 1 the loss vanishes iff $\delta \leq -g$. A minimum is $\leq -g$ iff one of its arguments is: either $\delta_c \leq -g$ (fitted), or $\max(u, -1) \leq -g$. Since $g < 1$ we have $-1 < -g$, so the latter holds iff $u \leq -g$, i.e. $\beta(2s - 1) \geq \delta_b + g$. For the escape region, $2s - 1 < 1$ gives $\beta(2s - 1) < \beta \leq \delta_b + g$ whenever $g \geq \beta - \delta_b$. Such a sample is not fitted either, since $g > 0 \geq -\delta_c$ (for $\beta > \delta_b \geq 0$). By Proposition 1 it then trains against j^* . For the absolute statistic and $g > g_{hi}$, $\delta \rightarrow \delta_b - \beta = -0.85$, and the only class that can satisfy $p_j > p_y + 0.85$ is j^* , so $K_\delta = \{y, j^*\}$. $\square$

Proposition 3. The batch mean and standard deviation of $ap + b$ are $a\bar{p} + b$ and $a \text{sd}(p)$, so every z_i and hence every $s_i = h(z_i)$ is unchanged. The empirical distribution of z has mean 0 and variance 1, and Cantelli’s inequality bounds the fraction with $z \geq k$ by $1/(1 + k^2)$. For increasing h , $s_i \geq h(k) \iff z_i \geq k$. $\square$

Corollary 2. With $s = \sigma(g/\tau)$, $2\sigma(x) - 1 = \tanh(x/2)$. By Proposition 2 a sample with $g > 0$ is rejected iff $f(g) = \beta \tanh(g/2\tau) - g - \delta_b \geq 0$. On $g \geq 0$, $\tanh$ is concave, so f is concave and its superlevel set $\{f \geq 0\}$ is an interval, possibly empty. The endpoints quoted in the text are computed on a grid of 2×10^5 points (`AbsoluteGapMargin.rejection_band`). `tests/test_mechanism.py` checks that the implemented loss has zero gradient exactly on this set, and positive gradient elsewhere, for $\tau \in \{0.1, 0.3\}$. $\square$

B The loss-mixture statistic

The loss-mixture statistic keeps a per-sample exponential moving average (momentum 0.5) of $-\log p_y$ over the steps at which each training example is seen. Every 100 steps, once half the training set has been seen, it fits one- and two-component Gaussian mixtures (20 EM iterations) to the *log* of these averages. The mixture is *active* only if BIC prefers two components and the high component’s mean loss is at least $\log C$, the loss of a uniform prediction. When inactive, $s_i = 1/2$ for every sample and the loss is exactly AS-Softmax. When active, s_i is the posterior of the high-loss component, and that component’s weight is logged as an estimate of the noise rate. This is the selection statistic of DivideMix (Li et al., 2020) expressed as a margin.

Design iteration before launch. The first design fitted raw losses and gated on BIC alone. A diagnostic on the training set with ground-truth flip labels (one CE epoch at 40% symmetric noise; no test accuracy involved) showed two problems. Raw-loss fits over-estimated the rate (0.59 at a true 0.40, precision 0.67), while log fits gave 0.39–0.40 with precision 0.82–0.87 and AUC 0.94–0.96. And on *clean* data BIC still preferred two components, with weight 0.44–0.52, because not-yet-fitted clean samples form a second mode. A rejected sample is never fitted, so that error would lock in. The two modes differ in location: mislabeled samples sat at a loss of about 3.5, above $\log 20 \approx 3.0$, while the unfitted clean mode was at 1.5 and falling. The chance-level anchor follows from that and is not a tuned threshold. In the main runs the mixture estimated the rate at 0.35 ± 0.01 (true 0.40) yet trained worse than the absolute rule (Table 1).

C Pre-registration record and earlier phases

The plan document and every decision rule live in the repository, timestamped by its version history. The negative-margin idea came out of an earlier, failed project on class-pair-structured margins for AS-Softmax. There, one two-seed run at 40% symmetric noise first showed the batch-relative

margin memorizing about a quarter as many corrupted labels as CE. Phase A was pre-registered to re-test that result on five seeds before anything was built on it. Table 4 lists every phase and Table 5 the 4-epoch results of Phases A–B’.

Table 4: Phases, the decision fixed before each, and the outcome.

phase	fixed in advance	outcome
A	5 seeds, 4 epochs, final epoch: M4 must memorize less than CE on every seed and beat it by > 2 sd; else stop	pass: +1.81 ($t = 4.4$), memorization 17% vs. 35%
B	$M4 \geq \text{small-loss} - 0.5$ and $\geq \max(\text{GCE}, \text{SCE})$	fail: GCE / small-loss / SCE all +1.7 to +2.3 ahead
B’	two named confounds (rejection timing; AS vs. CE base); one variant each; stop if the better is > 1 pt below small-loss	timing fixed: 67.41 vs. 67.80 (tie); CE base: 63.29
C	noise types and rates, β sweep, 8 epochs, noisy-held-out selection	best at sym 40%; fails clean (−3.8) and pair 40% (−4.6); noise-blindness diagnosed
C’	two self-calibrating statistics, P1–P4 (Table 3); P3 failure $\Rightarrow$ reframing write-up, no CIFAR-N	P3 fails for both; P4 passes for the absolute gap
E	instrumented re-runs (3 seeds, no new method or setting) for Section 5.4	42 runs reproduce Phase C exactly; AS gradient-share excess at epochs 1–4; clean lockout 23% / 13%; escape region $\leq 6\%$ of mislabeled samples, about 20% of the gradient

Table 5: Phases A–B’: 40% symmetric noise, 4 epochs, clean-test selection (*best*) and last epoch (*final*), 5 seeds. “M4, delayed” is the original schedule (rejection off for the first 30% of training, full strength only at the end); “M4, matched” uses small-loss’s schedule; “M4, CE base” makes the same rejection decisions as “matched” but trains kept samples with CE.

method	best	final	mem @final (%)
CE	67.46 $\pm$ 0.48	63.84 $\pm$ 0.50	34.7
AS-Softmax	67.70 $\pm$ 0.60	64.32 $\pm$ 0.40	39.4
GCE	68.28 $\pm$ 0.22	67.90 $\pm$ 0.53	9.3
SCE	68.18 $\pm$ 0.87	67.38 $\pm$ 0.54	9.1
small-loss (oracle)	68.09 $\pm$ 0.24	67.80 $\pm$ 0.27	12.4
M4, delayed	67.94 $\pm$ 0.34	65.65 $\pm$ 0.91	17.2
M4, $\beta = 0.5$, delayed	67.89 $\pm$ 0.42	64.68 $\pm$ 0.47	34.5
M4, matched	68.05 $\pm$ 0.53	67.41 $\pm$ 0.60	8.1
M4, CE base	65.29 $\pm$ 0.82	63.29 $\pm$ 0.95	12.4

D Reproducibility

Every run is one command, `python experiments/run.py --config <yaml>`, with the seed, noise rate and noise type as overrides.

- Configurations: `configs/phase_c/` (Phases C, C’) and `configs/final/` (Phases A–B’).

- Sweep drivers: `experiments/phase_c.py` and `experiments/mechanism.py`. Both also rebuild the summary tables.
- Figures: `experiments/make_paper_figures.py`.
- Tests: `python -m pytest tests/` (115 unit tests, including the closed-form checks of Section 3).

Hardware: one RTX 3080 Ti (12 GB), about 11 minutes per 8-epoch run.